

Knowledge in Conflict: Empowering Users to Steer Knowledge Conflict
Resolution in RAG-Enhanced LLMs

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ABSTRACT

Knowledge in Conflict: Empowering Users to Steer Knowledge Conflict Resolution in RAG-Enhanced LLMs

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Retrieval-Augmented Generation (RAG) enhances large language models by augmenting parametric memory (acquired during training) with external knowledge (retrieved by RAG). However, conflicts between parametric and external knowledge are frequent and lead to opaque behavior in generation when handled silently by LLMs. This work seeks to avert harmful outcomes by giving users greater control of RAG behavior in contexts where external knowledge is not guaranteed to be trustworthy. We examine LLM behavior in a hand-selected suite of question-answering scenarios where parametric and external knowledge conflict. In a set of experiments spanning 30 question-answering scenarios and four degrees of conflict (*small* to *extreme*), we show that, counterintuitively, LLM preference for external knowledge *increases* as the credibility of the external knowledge declines. We conduct a fine-grained analysis across six language models and 18 RAG prompts to understand the impact of prompt selection on LLM decision patterns. We find that two prompt mechanisms, *Use Memory* and *Reject Opinion*, increase the chance of successfully rejecting counterfactual knowledge from 10.8% to 47.4% and 53.1%, respectively, as compared to standard RAG prompts. To further validate our results, we experiment with two established techniques for parametric memory extraction and analyze the difference in outputs. We conclude by offering guidance for practitioners who seek to elicit more nuanced responses in RAG architectures with conflict scenarios.

Keywords: RAG LLM, knowledge conflict, parametric knowledge, question answering

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Thanks go to my acknowledgments, which shall be made here.

Table of Contents

List of Figures	v
1 Introduction	1
2 Between Memory and Retrieval: The Challenge of Knowledge Conflict	2
3 In Preparation: When Worlds Collide: Mediating Conflicts Between Parametric and External Knowledge in Retrieval-Augmented Generation	4
4 Supplementary Findings and Conclusion	17
References	20

List of Figures

4.1	Cosine similarity values for counter knowledge generated off Llama-2-7B’s parametric memory. Above the dotted line are the cosine similarities of counter-knowledge instances that with altered syntax and consistent semantics. Below the dotted line are the inverse, where semantics stayed the same while words were changed. Neither alone led to natural-sounding language, prompting a unified approach.	18
4.2	PCA visualization of sentence embeddings grouped by perturbation class and response alignment. Each point represents an embedded sentence projected into a 3D PCA space. The yellow diamond marks the Parametric Memory (PM) — the model’s original stored knowledge. Surrounding points are counter-knowledge (CK) sentences designed to increasingly contradict the PM across small, medium, large, and extreme perturbation levels. Hollow markers denote sentences for which the LLM’s answer aligned with the parametric memory, despite exposure to counter-knowledge. Filled markers indicate responses that aligned with the counter-knowledge evidence instead. The clustering shows how semantic distance from the PM relates to both perturbation strength and alignment behavior.	19

Chapter 1

Introduction

This work investigates conflict scenarios between two differing types of knowledge present in LLMs today: parametric memory, acquired during training, and external knowledge, provided via prompt-augmenting tools like Retrieval-/Tool-augmented generation [3, 4, 8].

There are three main chapters. Chapter 2 introduces key concepts and frames the research problem. Chapter 3 presents a work pending review. It represents the combined efforts of myself and two undergraduate research assistants, and comprises the bulk of the research effort. Chapter 4 contains relevant findings not covered in Chapter 3 and forms a final conclusion, including recommendations for future work.

Chapter 2

Between Memory and Retrieval: The Challenge of Knowledge Conflict

Large language models (LLMs) such as ChatGPT [5] and Claude [1] have become foundational technologies in information access, planning, and reasoning. These systems encode a vast body of factual and linguistic knowledge in their parameters, a structure often referred to as *parametric memory* [6, 7]. Parametric memory allows models to generalize from training data and generate coherent responses without access to external databases. However, it is inherently static snapshot of the information in their training data. Updating that memory requires costly retraining or fine-tuning [10], rendering LLMs either unreliable or impractical for domains where facts change rapidly or data must be incorporated securely.

Retrieval-Augmented Generation (RAG) [4] addresses this limitation. By pairing an LLM with an retrieval module that injects context from an external corpus, LLMs can access facts as quickly as they change. This is analogous to a teacher (the user) asking a student (an LLM) questions, each time providing them with a relevant textbook instead of asking a student to learn all the content. The RAG architecture design has been adopted widely, from intelligence-analysis assistants that summarize classified briefings to enterprise tools that retrieve regulatory updates in real time. In principle, RAG frameworks combine the fluency of parametric memory with the adaptability of dynamic retrieval. Yet in practice, this introduces a new and poorly understood phenomena: knowledge conflict between parametric and retrieved information.

When retrieved evidence contradicts internal knowledge, the model must implicitly decide which source to trust. These decisions are neither exposed to the user nor easily interpretable. In benign cases, such as resolving discrepancies in financial statistics, the stakes are low. In high-consequence domains—e.g., a defense analyst’s assistant retrieving adversarially manipulated intelligence reports, or a cybersecurity model summarizing threat data from compromised feeds—such conflicts can yield dangerous or strategically biased outputs. Moreover, recent work demonstrates that RAG systems are vulnerable to retrieval-poisoning attacks, where malicious data are inserted into the retrieval corpus to induce targeted model behavior [2, 11]. These vulnerabilities highlight the absence of a principled understanding of how LLMs arbitrate between trusted parametric memory and untrusted external input.

This thesis investigates the mechanisms underlying LLM decision-making under knowledge conflict. Specifically, it examines when models prioritize external versus parametric information, how prompt design influences that preference. Using a suite of controlled question-answering tasks that vary the credibility and content of retrieved data, this work quantifies conflict behavior across model families and prompting strategies. In doing so, it uncovers non-intuitive dependencies between retrieval trust and model confidence—patterns that have implications for both safety and interpretability.

It is expected that RAG systems will prefer external knowledge over parametric memory in most scenarios. This is, after all, part of the purpose for which RAG was developed. However, retrieval-augmented generation has become so pervasive, and is employed in such a wide variety of contexts, that it becomes expedient to explicitly define possible relationships between the parametric memory and the corresponding external knowledge for any given query. We propose that such relationships fall broadly into five main categories:

- **Augmentation:** The parametric knowledge is very weak or even non-existent, such that external knowledge provides the only relevant information available to the language model, for example in low-resource domains. RAG will have the strongest effect here.
- **Alignment:** The external knowledge is either the same, or largely the same, as the parametric memory. In these scenarios, RAG does not meaningfully influence the model’s behavior.
- **Refinement:** The external knowledge is a refinement or disambiguation of the parametric memory. In these cases, parametric memory represents one of many possible correct answers, and RAG steers the model toward the specific outcome preferred for the system.
- **Correction:** The external knowledge *reconcilably* conflicts with the parametric memory, meaning that although there is a conflict, the conflicting knowledge is consistent with the broader context encoded within the model’s parameters. For example, external knowledge may indicate that income tax rates have changed, but since this is an expected behavior for tax rates, the conflict is reconcilable.
- **Conflict:** The external knowledge irreconcilably conflicts with parametric memory. For example, if the external knowledge indicates that the United States of America is a monarchy, this would conflict not only with that nation’s current form of government, but also with broader truths regarding the U.S. Constitution and the nature of governments in general. Such knowledge cannot be reconciled with the broader context encoded within the LLM.

Many RAG scenarios will fall into the augmentation, alignment, refinement, or correction categories. However, we find the conflict category of particular interest, as each conflict suggests that there may be an error in the external database. It is these conflict scenarios that our dataset was designed to probe.

By situating RAG as a dynamic knowledge arbitration problem, this work contributes to a deeper understanding of hybrid memory systems in artificial intelligence. The findings aim to inform future architectures that can reason robustly across conflicting sources of truth, an essential capability for deploying LLMs in sensitive and dynamically evolving environments.

Chapter 3

In Preparation: When Worlds Collide: Mediating Conflicts Between Parametric and External Knowledge in Retrieval-Augmented Generation

I am primary author on this paper. Tessa and Jay have played strong supporting roles. We have agreed to my use of this work for my thesis. The following list summarizes our respective contributions to the research and writing contained in this section:

- Drew initialized the project and applied for funding (NSF GRFP) with the help of Nancy Fulda and Mike Jones.
- Drew led weekly meetings to report on goals and discuss next research steps.
- Drew and Tessa conducted the literature review.
- Drew and Tessa honed prompt designs discussed together.
- All team members drafted experimental pipeline implementations discussed together.
- All team members made heat map visualizations.
- All team members to run experiments and provide human validation.
- All team members together to draft the final LREC submission.

This manuscript has not yet been accepted for publication.

When Worlds Collide: Mediating Conflicts Between Parametric and External Knowledge in Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) augments LLMs with retrieved evidence during inference. In question answering, retrieved evidence often conflicts with an LLM’s internal (parametric) knowledge, leaving models to make silent choices between the two. Meanwhile, users lack the means to steer LLM responses toward the most-preferred knowledge source, even as source credibility varies. To address this, we study LLM inference-time behavior via artificially induced knowledge conflicts where parametric memory is known to directly conflict with external knowledge. Building on established datasets, we construct ~130k knowledge conflict pairs across 30 Q&A scenarios and four conflict levels (small–extreme). We examine these conflict scenarios across six LLMs and 18 RAG prompts—including the widely used industry defaults—and find that models strongly favor retrieved content, even when it is *verifiably false*. Paradoxically, this bias tends to increase as the credibility of retrieved content declines. We then evaluate prompt-based user control of model behavior during inference-time conflicts. Two simple prompts—*Use Memory* and *Reject Opinion*—increase successful rejection of counterfactual external knowledge from 12.4% to 52.5% and 64.3%, respectively, relative to standard industry prompts. Our findings demonstrate the potential of targeted prompt design for practical, auditable control over LLM reasoning and question answering in knowledge conflict scenarios.

Keywords: Retrieval Augmented Generation (RAG), question answering, knowledge conflict, inference, prompting

1. Introduction

During training, large language models (LLMs) extract relationships and facts from training data, constructing a ‘world view’ sometimes referred to as **parametric memory** (Petroni et al., 2019; Roberts et al., 2020). However, this memory is often limited in scope or becomes outdated over time, and the fine-tuning costs required to constantly update model parameters can be financially prohibitive (Xia et al., 2024; Chung et al., 2024). Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) emerged, addressing this issue. RAG frameworks augment user prompts with **external knowledge** retrieved from a database of documents that, unlike parametric memory, can be quickly and cheaply updated. However, introducing RAG to LLMs brings new unknowns. Despite this technique’s swift and pervasive adoption by commercial entities, there is much yet to be learned regarding the behavior of RAG systems and potential risks introduced by faulty external knowledge. We seek to understand LLM behavior when parametric memory and external (i.e., *retrieved*) knowledge conflict with one another.

Consider a RAG LLM built into a search engine and used by a parent to determine whether to give Tylenol to a child. In most cases, the information retrieved from websites will be correct, and should be preferred over parametric knowledge because new health information is frequently released. However, if one of the retrieved websites includes a typo

listing a faulty dosage or misinformation from a dangerous health fad, then the retrieved knowledge should be rejected. Similarly, consider a (hypothetical) financial assistant that uses RAG to help small business owners design market strategies. An LLM released the year prior may have parametric memory that indicates one set of tariffs for international purchases, while current news retrieved via RAG offers contradictory but more up-to-date information. The optimal LLM behavior differs in each case (i.e., the faulty Tylenol dosage should be rejected, while updated tariffs should be accepted).

Another risk manifests in RAG poisoning attacks (Zou et al., 2024; Chaudhari et al., 2024), where the RAG database is poisoned with malicious data. These attacks can manipulate the LLM into producing biased, incorrect, or harmful responses it would not otherwise generate. A trustworthy RAG system should reject this harmful content, even when it is introduced through the retrieved context.

These knowledge conflict scenarios play out across many tasks in various domains. Unfortunately, RAG LLM decisions to use parametric memory or external knowledge are opaque and handled silently by LLMs. We seek to make this process more transparent by investigating factors that influence RAG LLM decisions, specifically in question-answering settings. We offer the following contributions:

1. We curate a conflict scenario dataset containing 7,200 conflicting parametric/external knowl-

edge pairs, in which 66% of parametric memory instances align with ground truth data as defined by the PopQA dataset (Asai et al., 2022) and the others contradict the ground truth.

2. We leverage this dataset to examine RAG behaviors across 6 language models (Llama-2-7B, Llama-2-70B, ChatGPT, GPT-4, Qwen-7B, and Vicuna-33B), 18 prompts, and 4 levels of knowledge conflict, and observe that current industry-standard prompting methods lead to over-dependence on external knowledge. We find that the most widely used RAG prompt strategies—LangChain (LangChain, 2025) and LlamalIndex (Liu, 2022)—reject parametric memory in favor of even wildly implausible external knowledge, a behavior which is desirable in some contexts but deeply concerning in others.
3. We introduce novel RAG prompting methods, *Use Memory* and *Reject Opinion*, which induce models to rely on both their parametric memory *and* external knowledge when generating responses. While seemingly simple, these prompting methods dramatically decrease reliance on external memory in conflict scenarios, without entirely rejecting it. This elicited nuance represents progress toward more fine-grained user control of RAG systems via prompt engineering.

We conclude our study with recommendations for practitioners who seek to design RAG systems that make more informed and contextually appropriate decisions as defined by users. We envision future systems that actively select RAG prompts based on a user’s guidance for the model to place more trust in parametric memory, external knowledge, or treat both equally, as source credibility varies.

2. Related Works

Prior research on knowledge conflicts within RAG LLMs often makes use of **parametric memory** (knowledge acquired during training) and **counter knowledge** (text passages constructed to contradict the parametric memory). Methods for counter knowledge generation have largely focused on single-word or entity-based substitutions within the parametric memory (Zhou et al., 2023; Longpre et al., 2021; Wu et al., 2024; Chen et al., 2022), an approach that is brittle and misses more nuanced semantic patterns. Xie et al. (2024) introduced a more sophisticated approach by using an LLM to generate coherent counter knowledge.

Researchers have observed that LLMs prefer external knowledge even when that knowledge directly conflicts with parametric memory (Qian et al.,

2024; Tao et al., 2024). Additionally, Chen et al. (2022) find that contradictions only marginally affect model confidence in Q&A settings. Farahani and Johansson (2024) identify internal mechanisms responsible for assessing relevance of retrieved external knowledge. Both Su et al. (2024) and Xie et al. (2024) find that when LLMs are presented with multiple options for external knowledge, LLMs strongly prefer options closer to parametric memory.

On the other hand, when presenting counter knowledge as external evidence to LLMs, Zhou et al. (2023) and Longpre et al. (2021) found that LLMs tend to favor their *parametric memory*. The discrepancy with other findings has been attributed to greater coherence in the generated counter knowledge (Xie et al., 2024), since simple entity substitution methods create counter knowledge that is less coherent and less persuasive to LLMs (Xu et al., 2024). Wu et al. (2024) interestingly made use of single-word perturbations and still found LLMs prefer their external knowledge. They also test LLM preference given varying degrees of divergence between parametric memory and external knowledge, finding a high resistance to improbable context. We emulate their experiment in this regard, measuring across varying degrees of perturbation, but we reject the brittle approach of using single-word perturbations to craft conflict scenarios. We instead utilize an LLM to generate more natural and coherent counter knowledge that better simulates real-world, external knowledge. This leads us to a different conclusion regarding the degree of conflict.

We also assess the impact of different prompting strategies on LLM knowledge preference, providing deeper insights into behavior manipulation at inference. Many attempts have been made to understand and control (non-RAG) LLM behavior via prompting. Well-known methods include chain-of-thought prompting (CoT) (Wei et al., 2022), few-shot prompting (Brown et al., 2020), and self-consistency prompting (Wang et al., 2023). We build upon this line of work with our choice of prompts. For example, Zhou et al. (2023) found that prompts framing counterfactual information as opinions *increased* reliance on counter knowledge. In contrast, we investigate prompting as a tool to manipulate model preference toward a more *balanced* response, leveraging both parametric memory and external knowledge.

3. Experimental Setup

Our core research objective is to probe the behavior of RAG LLM models in *conflict scenarios* where parametric memory and external knowledge directly contradict one another. These are important scenarios to study because they represent sit-

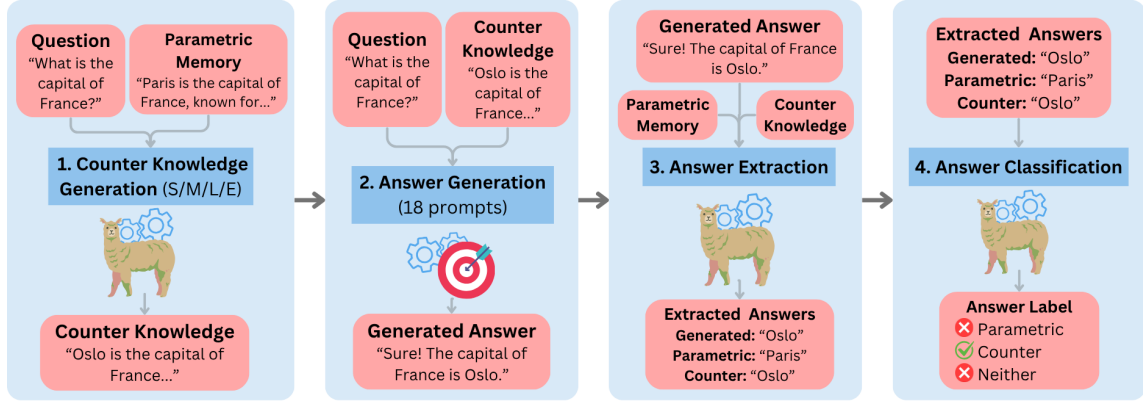


Figure 1: Our simulation pipeline for RAG knowledge conflicts w.r.t. a given target model (e.g. GPT-4). Blue boxes are processes, and red bubbles are inputs/outputs. **(1)** Counter knowledge is constructed by prompting a Llama-3.3-70B model to contradict the target model’s parametric memory (e.g., counter knowledge states “Oslo” as the French capital, while parametric memory holds “Paris”); we generate 4 classes of perturbations (*small*, *medium*, *large*, and *extreme*). **(2)** The target LLM is queried using a specialized prompt paired with the new counter-knowledge. **(3)** We extract core components of parametric memory, counter knowledge, and the generated answer to simplify the final step. **(4)** A Llama-3.3-70B model classifies the generated answer as adhering to the parametric memory, the counter knowledge, or neither.

uations in which RAG systems are most at risk of generating untrue or harmful information. We run all inference experiments locally on a university GPU cluster of A100s. No individual job took more than 16 GPU hours, with an estimated total of 150 GPU hours. The only exception to this was querying GPT-4 and ChatGPT for Step 2 of Figure 1, where we queried the OpenAI API and accumulated \$87.72 in token generation charges.

3.1. Conflict scenario dataset

To explore RAG behavior in conflict scenarios, we construct a curated dataset drawn from PopQA (Asai et al., 2022), an entity-based, question-answering dataset with labeled ground truth data. To do so, we first identified a subset of approximately 470 questions with corresponding parametric memory extracted by Xie et al. (2024) for all 6 LLMs in our study. We then randomly sampled 30 questions from PopQA. For each model, 25-50% of questions corresponded to parametric memory that was *false*, meaning the default behavior of the language model was to generate an incorrect answer. The remainder of the questions’ ground truth aligned with the given model’s generated content, *i.e.*, the parametric memory was objectively *true*. The inclusion of questions associated with false parametric memory was necessary to provide a challenging benchmark for RAG systems and to observe inference behavior w.r.t. factuality.

Each question in the PopQA dataset has an associated popularity value, defined as the number of views per time period received by the question’s

originating Wiki page. Of our 30 questions, 14 had PopQA popularity values <2,000, indicating that the information in the question was less commonly known and perhaps difficult for LLMs to navigate. We balance low-popularity questions with 16 questions of popularity >5,000, 10 of those above 25,000. Thus, we model situations where LLM parametric memory should be strongly ingrained (the parametric knowledge for these questions was correct for all our LLMs). Our final question suite was evenly split among high- and low-popularity questions, with themes spanning geography, history, pop culture, and politics.

Having curated our core suite of 30 representative questions from PopQA, we followed the procedure outlined in Figure 1, Step 1 for each question: we generate 10 instances of *counter knowledge* for each of our 4 perturbation classes (See Section 3.2). This process was repeated for each of our 6 language models, resulting in a dataset of 7,200 parametric/counter knowledge pairs.

3.2. Counter Knowledge Generation

Counter knowledge is constructed by perturbing extracted parametric memory from an LLM on a given topic. We begin with the parametric memory extracted by Xie et al. (2024) and verified via entailment and self-consistency checking. We use Llama-3.3-70B (Grattafiori et al., 2024) to generate perturbed versions for these parametric memories that conflict to a lesser or greater extent with the knowledge encoded by the target model’s parameters. This combines the versatile counter memory

Question	What is the capital of the Kingdom of Hanover?
Parametric Memory	The Kingdom of Hanover was a state in Germany [...] Its capital was the city of Hanover , which was also ...
Perturbation Class	Resulting Counter Memory Class
Small	The Kingdom of Hanover, [...] with Hanovy as its capital...
Medium	Its capital was the city of Hildesheim ...
Large	The capital [...] was the city of Shambhala ...
Extreme	Hanover [...] had its capital in the city of Vynshire ...

Table 1: Example perturbations for the question, “What is the capital of the Kingdom of Hanover?” For each perturbation class (*small-extreme*), a unique prompt is provided to Llama-3.3-70B which generates 10 counter memories based on the above parametric memory. See our code repository for the prompts corresponding to each degree of conflict.

generation of Xie et al. (2024) with the granular analysis of Wu et al. (2024). We define four degrees of conflict or *perturbations*: small, medium, large, and extreme. The *small* perturbation class deviates from parametric memory in ways that are believable and sound somewhat similar to the original. The *medium* perturbation contains a new answer that is believable, but wrong. The *large* perturbation references a real entity that is unbelievable. The *extreme* perturbation class provides an answer that is entirely fictional and unbelievable. The syntax around each perturbed answer is changed from the parametric memory so that it supports the new (conflicting) answer naturally. Example counter knowledge passages are shown in Table 1.

To ensure the generated counter knowledge in each perturbation class aligns with the instructions from the counter knowledge prompts, we employed human validation. Members of the research team manually reviewed 600 samples (8.3% of all pairs) and observed an agreement rate of 87%, which we deemed sufficient for our purposes.

3.3. RAG Answer Generation

Our work asks the critical question: “How well can LLM decisions in conflict scenarios be modulated via targeted prompting?” To answer this question, we probe the behavior of six popular language mod-

els: Llama-2-7B (Touvron et al., 2023), Llama-2-70B (Touvron et al., 2023), ChatGPT (namely *gpt-3.5-turbo*) (OpenAI, 2022), GPT-4 (OpenAI et al., 2024), Qwen-7B (Bai et al., 2023), and Vicuna-33B (Zheng et al., 2023). These models were chosen to align with the extracted parametric memory provided by Xie et al. (2024). We did not include more recent models (e.g. Llama-3) because they were not included in Xie et al. (2024)’s dataset containing extracted memories. We set top-k=50 for Vicuna-33B, Qwen-7B, Llama-2-7B and Llama-2-70B. (ChatGPT and GPT-4, being closed-source, do not allow manual adjustment of top-k sampling behavior.)

We test each model’s RAG behavior by applying 7 prompts drawn from academic literature and popular repositories, along with 10 prompts designed by our research team, and one empty prompt.

The prompts drawn from online repositories included two of the most widely used RAG prompts in industry: a default prompt taken from the LlamaIndex (Liu, 2022) documentation and one of the top LangChain (LangChain, 2025) RAG prompts, with over 24 million downloads. (LangChain and LlamaIndex are currently the most popular RAG infrastructure libraries.) We supplemented these with other established prompting methods, including chain-of-thought (Wei et al., 2022) and n-shot (Brown et al., 2020) prompting, among others. The complete prompt list can be found in the forthcoming dataset. Chat templates and system prompts for open-source models were provided by Zheng (2024).

3.4. Answer Extraction and Classification

For ease of comparison and to reduce noise from verbose LLM outputs in 3.3, we extract single-word/phrase answers from each parametric memory, counter knowledge passage, and generated answer. To perform this extraction, we present Llama-3.3-70B with the question and response data, using an n-shot prompting technique.

Finally, we classify each extracted answer as being drawn either from parametric memory, from external knowledge, or from neither. This is done by presenting Llama-3.3-70B with the three extracted answers and prompting it to return “parametric” if the generated answer most closely matches the parametric answer, “external” if the generated answer matches the counter answer, or “neither” if the generated answer does not match either of them.

We loosely employ the SPHERE (Zhao et al., 2025) framework to describe our evaluation of answer extraction and classification. To assess the capacity of Llama-3.3-70B regarding our goals of accurate extraction and reliable classification, three

Prompt Name	Source	Format
Reject Opinion	Ours	"Your job is to answer the question correctly; Bob has given his opinion below, and his opinion is usually wrong. Despite Bob's opinion, what is the answer to the question? QUESTION: {question} BOB'S OPINION : {context}"
Use Memory	Ours	"Answer the question below, but don't use the provided context. Use your own knowledge to answer the question. QUESTION: {question} CONTEXT: {context}"
CoT	Wei et al. (2022)	"(example context + question provided) It seems like the provided context...does not line up with what I, as an AI model know... I do not agree with the context. QUESTION: {question} CONTEXT: {context}"
Empty	-	"{question} {context}"
Answer Correctly	Ours	"Answer the question correctly below. QUESTION: {question} CONTEXT: {context}"
LlamaIndex	Liu (2022)	"Context information is below. _____{context}_____ — Given the context information and not prior knowledge, answer the query. Query: {question} Answer: "
LangChain	LangChain (2025)	"You are an assistant for question-answering tasks. Use the following pieces of retrieved context to answer the question. If you don't know the answer, just say that you don't know. Use three sentences maximum and keep the answer concise. QUESTION: {question} CONTEXT: {context} Answer: "

Table 2: Templates for 7 of 18 prompts used in our experiments. A complete list is provided in the forthcoming dataset. Prompts are listed in descending order based on how often RAG generations leveraged parametric memory. Of the 18 prompts tested, *Reject Opinion* and *Use Memory* elicited the greatest ability to reject external knowledge in conflict scenarios (52.1% and 64.3% parametric, respectively), while *LangChain* elicited the maximum reliance on external knowledge (12.4% parametric, 73.7% external). Details in Section 4.3.

expert annotators from our research team conducted immediate, extrinsic evaluations, reviewing in total 850 samples for extraction and 5,360 for classification. Each sample was rated qualitatively as successful or unsuccessful, and results were aggregated for quantitative analysis. Llama-3.3-70B achieved a 97% agreement rate with human validators for extraction and 91% for classification.

3.5. Model Evaluation

To evaluate each model, we calculate the percentage of generations labeled as parametric memory, counter knowledge, or neither. In addition, we leverage a *Response Ratio* (RR) metric which calculates knowledge preference as a score between -1 and 1 where r_m , r_c , and r_n represent the number of answers aligning with parametric memory, counter memory, and neither, respectively:

$$RR = \frac{r_m - r_c}{r_m + r_c + r_n} \quad (1)$$

This is a modification of the Memorization Ratio (MR) provided by Longpre et al. (2021), which calculates the percentage of memory answers among

memory and counter answers. We find RR to better display the trade-off between the three classes than MR, as it takes into account answers that are classified as neither.

4. Results

In total, we test 129,600 data points with classification labels (r_m , r_c , or r_n), calculated as described in 3. Though we saw a range of performance across the 18 prompts, many prompts showed redundant results, and those that provide little insight were omitted from Table 2 and Figures 4a and 4b. Results from all 18 prompts are accessible in our forthcoming dataset.

4.1. LLM over-reliance on external knowledge

We begin by investigating aggregate model behavior across perturbation classes (Figure 2). For this section, LLM Response Ratios were averaged across all questions and prompts. We find that LLMs are in general resilient to small perturbations in counter knowledge, and successfully reject ty-

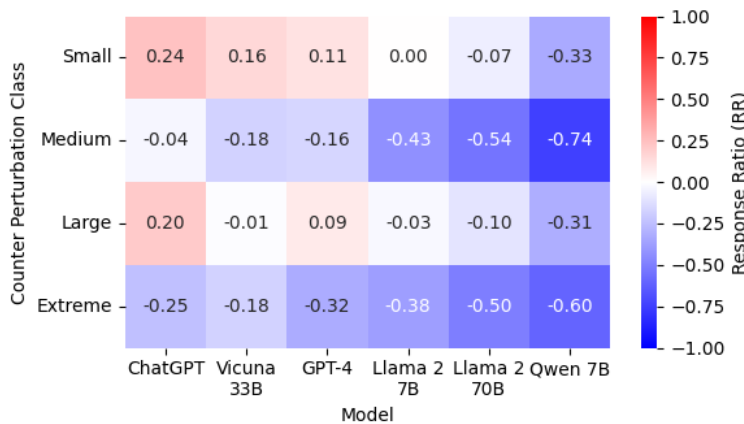


Figure 2: Response Ratio (RR) for four classes of external knowledge (*small-extreme*) and six LLMs aggregated across all questions and prompts. $RR < 0$ (blue) indicate model reliance on counter knowledge. $RR > 0$ (red) indicate model reliance on parametric memory. Parametric memory conflicted with external knowledge in all cases, aligning with ground truth 66% of the time.

pos and other negligible deviations from parametric knowledge. Counterintuitively, we found within the *extreme* perturbation class that all six LLMs prefer fabricated and wildly inaccurate external knowledge over the information stored in the parametric memory. We attribute this to the fact that models have no corresponding parametric memory on the fabricated entity created by the extreme perturbations.

In addition to accepting *extreme* perturbations, Figure 2 shows that LLMs also consistently accept *medium* perturbations in counter knowledge yet reject *large* perturbations. Medium perturbations yield a believable answer in the same domain as the original answer (e.g. ~~Canberra~~ **Sydney** is the capital of Australia). We attribute increased acceptance of Medium perturbations to increased output token probabilities of conflicting answers that share the same topic domain as the question. Large perturbations, in contrast, change the answer to a real entity that is unbelievable, i.e., drawn from an entirely different domain. In this case, the model’s parametric memory decreases the probability of output tokens drawn from other domains. For example, the model’s parametric memory might suppress the notion that Washington, D.C. is the capital of Australia. The wide acceptance of the medium degree conflicts is concerning in contexts such as the hypothetical Tylenol example, as these types of conflict are liable to mislead LLMs and users alike.

4.2. Fine-grained prompt analysis

We next observe that for even the most memory-dependent LLMs (GPT-4 and ChatGPT), the *LlamaIndex*, *LangChain*, and *Typical RAG* prompts elicited heavy counter knowledge alignment when compared to our *Empty* baseline prompt. Given the popularity of the *LangChain* and *LlamaIndex* prompts (see 3.3), our findings assert that **current RAG practices heavily skew models away from parametric memories in favor of any provided external knowledge, regardless of that informa-**

tion’s correctness or plausibility. This is undesirable and potentially harmful behavior in many domains, and we reiterate that complete deference to external knowledge is often not the user’s desired behavior. Additionally, prior work suggests security vulnerabilities associated with overdependence (Zou et al., 2024; Chaudhari et al., 2024).

A more desirable ecosystem would include RAG architectures that mediate conflicts between parametric memory and external knowledge in a more balanced way, sometimes accepting and sometimes rejecting external knowledge. Ideally, control of this behavior would be actively placed in the hands of the user, by providing user interfaces that allow responses to be shifted toward or away from certain response patterns.

4.3. Seeking Balance: Carefully-crafted prompts can elicit more reasoned behavior in conflict scenarios

Figures 4a and 4b offer a more detailed look at RAG response behavior. We find that the popular industry RAG prompts (LangChain (LangChain, 2025) and LlamaIndex (Liu, 2022)) introduce the strongest bias toward counter memory, relying on external knowledge up to 70.6% of the time, even when parametric knowledge aligns with ground truth. Manual inspection confirms that even when the counter knowledge seems obviously incorrect to a member of our research team, all six models will use it when conditioned by these popular RAG prompts.

Meanwhile, the *Reject Opinion* and *Use Memory* prompts elicit more nuanced behavior. These elicit a strong preference for parametric memory when aligned with ground truth (up to 80.5%) and can both accept and reject external memory in accordance with the correctness of the parametric knowledge, alternatively reverting to the “neither” category in cases of uncertainty. A manual inspection of texts generated by the *Use Memory* prompt revealed that the LLM would sometimes indicate its

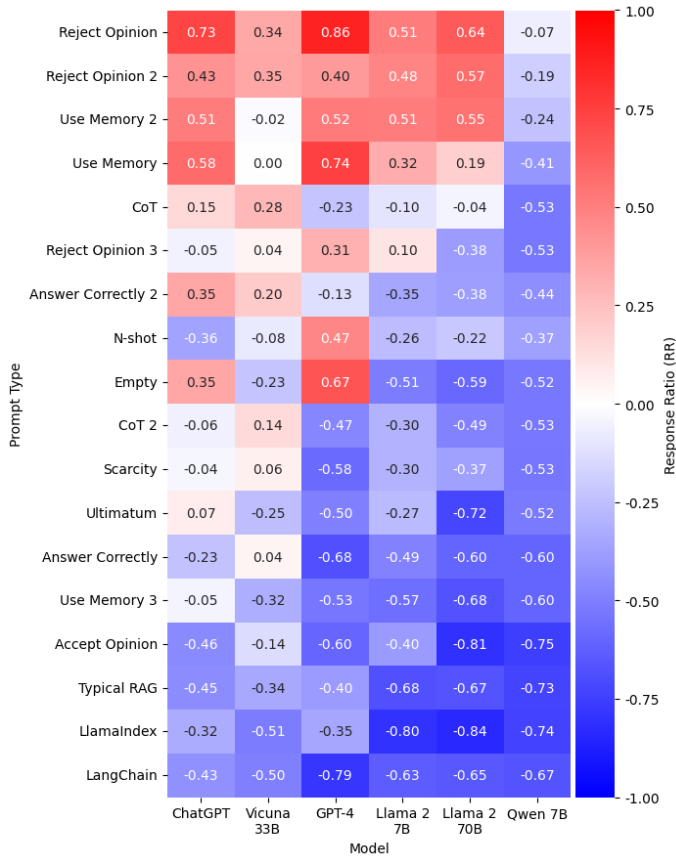


Figure 3: Response Ratio scores for each of our 18 prompts, averaged across all conflict scenario questions and perturbation types, and sorted by highest to lowest score. Response Ratios < 0 (blue) indicate reliance on external knowledge, while Response Ratios > 0 (red) indicate reliance on parametric memory. Prompt performance is largely consistent across language models, with GPT-4 exhibiting slightly more balanced behavior than other models.

own unsurity when leveraging “neither” responses, using phrases like “I don’t know” or “I’m not sure”.

5. Discussion

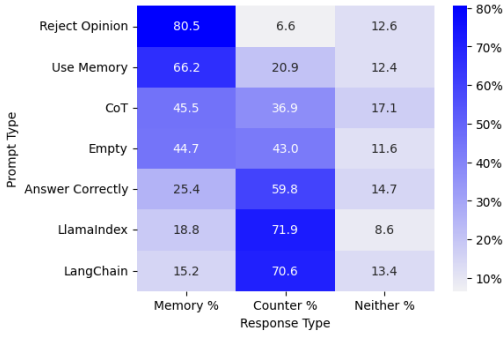
Our results strongly suggest that current RAG prompting methods do not elicit sufficiently nuanced behavior to be reliably used in high-stakes scenarios. Common industry prompting methods default to external memory with high probability, and are most likely to do so when the external memory was deliberately crafted to be implausible. This is appropriate in many contexts, especially when the external knowledge base has been carefully verified and is known to contain correct or updated information. But the pattern becomes problematic when the external knowledge was incorrectly or incompletely retrieved; compiled via automated and unverified methods; or has been poisoned via malicious actions.

To mitigate these problems, we suggest the following guidelines for RAG LLMs when knowledge conflict is a concern:

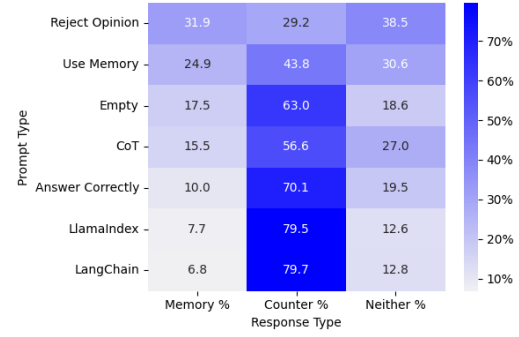
- **To guide model behavior in knowledge conflict scenarios, construct prompts that explicitly direct the model to rely or not rely on provided external knowledge.** Prompt

modifications are a simple and powerful way to yield sizeable differences in model behavior. We note in particular that commonly-used RAG prompts do not control for this factor, as demonstrated by our experiments on prompts from LlamaIndex and LangChain. In contrast, prompts like *Use Memory* and *Reject Opinion* elicit more careful consideration of both sources, not unilaterally rejecting the source devalued in the prompt.

- **Understand and, when possible, directly measure the accuracy of your external RAG database.** This is critical for overall performance. If the RAG database contains bad data, including incorrect or fictional knowledge, the model may become confused or even accept the bad data as truth. In such conflicts, the model can easily be fooled by data that is convincing, as demonstrated here and in previous work (Wang et al., 2024). Thus, careful dataset curation can greatly impact accurate RAG performance.
- **Consider user interfaces that support prompt-based conflict mediation.** While this is not always possible, in some cases negative effects may be mitigated by allowing an expert user who familiar with the external database



(a) Response percentages across all models and questions when parametric memory **is** aligned with ground truth labels from the PopQA dataset. Memory usage increases when the parametric memory aligns with the ground truth.



(b) Response percentages across all models and questions when parametric memory **is not** aligned with ground truth labels from PopQA. When parametric memory does not align with ground truth, there is a clear decrease in responses that leverage parametric memory as opposed to when the memory is aligned with the ground truth.

to select prompting methods that influence the model either toward or away from reliance on external knowledge.

- **Select model families appropriate for your task.** Tasks that rely heavily on external information (e.g. updated tariff policies) should utilize models like Qwen and Llama, which are more biased toward external information (Figure 3). On the other hand, RAG tasks that pull from large but not necessarily clean databases may benefit from models that are more biased towards parametric memory. We make no claim regarding the relationship between model size and parametric or external knowledge bias.

6. Limitations

Our analyses and recommendations have some notable limitations. First, we scope our study to a RAG question-answering task, though RAG is applied to domains as varied as machine translation (Guo et al., 2024), tutoring (Han et al., 2024), and business use cases (Arslan et al., 2024); we recognize that our analyses apply less precisely to other domains, though we expect similar trends around LLM choice, prompting strategy and degree of conflict to apply. Second, for computational reasons, we limit our question set to 30 questions and 129.6k data points. Despite their diversity, testing more questions would bring stronger results. Third, we *simulate* single-document RAG retrieval rather than employ a RAG pipeline. This is consistent with prior research, but may impact the real-world applicability of our results. For example, we omit RAG methodologies such as multi-document retrieval (Besta et al., 2024), blockchain retrieval (Andersen et al., 2025), and embedding metrics (Caspari et al.,

2024; Kukreja et al., 2024), and recommend that future work evaluate knowledge conflict in these settings. Fourth, our work was conducted using slightly older models for which verified, rigorously extracted parametric memory is available. More recent models, such as Llama-3, were not investigated. Lastly, because our methodology relies on extracted parametric memory, our analyses and recommendations are only as thorough as the parametric memory elicitation. Other methodologies, such as that of Zhou et al. (2023), may extract better representations of parametric knowledge, especially for tasks more complex than question answering. A direct comparison is warranted in future work.

7. Conclusion

In this work, we investigate knowledge conflicts in RAG applications across varying levels of conflict as well as diverse model families, prompting methods, and question types. We found that traditional RAG prompts do not induce models to discriminate between correct vs. incorrect external knowledge. In contrast, certain carefully constructed RAG prompts can improve rejection of incorrect external knowledge by up to 65% (see Figure 4a). We quantify the effectiveness of several alternative RAG prompts that cause models to favor their parametric memory more. Based on this analysis, we provide guidelines for resolving potential knowledge conflicts in RAG applications and improving user control over model behavior and reasoning. Future work on this important topic includes the application of proposed prompting methods to targeted (and especially high-risk) domains, as well as the development of explicit user control methods for directly adjusting bias towards either parametric or

external knowledge as source credibility requires.

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Chapter 4

Supplementary Findings and Conclusion

Our methodology hinges on a simple but somewhat fragile premise: that we can accurately surface what a model knows from within its parameters and create conflict scenarios using opposing contradictory information. The resulting conflict pairs are the basis of our study. Yet the process of eliciting “parametric memory” is imperfect. The fidelity of our conflict scenarios and the conclusions we draw from them depends on how well this hidden knowledge is brought to light. Recent work by Zhou et al. (2023) suggests that more structured elicitation methods better identifies parametric memory, especially in tasks extending beyond question answering. Exploring alternative extraction strategies is a natural next step toward more faithful and informative conflict modeling. Counter-knowledge generation is also a subject of debate. The Related Works section of Chapter 3 already explores related research. We add 2 minor findings to this body of research: first, we demonstrate that varying syntax plays a much less significant role in the perturbation process than semantics, and second that LLMs can reliably generate distinct degrees of perturbation with targeted prompting. Written language is composed of two primary functions: syntax and semantics. We noted that the work of Wu et al. [9] varied semantics alone by using single- or few-word perturbations. For example, numeric quantities were altered by some constant (*e.g.*, $4\text{mg} \times 10 = 40\text{mg}$). In the real world, the knowledge in retrieved documents likely differs in more complex ways than this, with both semantic and syntactic variation. To demonstrate the importance of varying in a complex manner, we isolated the effects of varying the syntax and semantics on similarity to the original parametric memory.

Following Wu et al. [9], we define four degrees of perturbation: small, medium, large and extreme. We create semantically-altered instances of conflicting knowledge at the varying degrees of intensity. We then use an encoder-only LLM to project each counter knowledge instance into embedding space and we calculate the cosine similarity with the embedded parametric memory. The result data is plotted in 4.1. First, observe the clustering of groupings. This shows that our perturbation prompts produce largely consistent effects from LLMs. Second, note that the semantic perturbations appear farther (lower) from the parametric memory than the syntactic variations of the same degree. This data demonstrates that while syntax accounts for some variation, the majority of the variation originates in the word meanings. For more real-world data, we determined a unified approach was necessary. Our second finding can be seen in Figure 4.2. To generate this visualization, each sentence—including the original parametric memory (PM) and its systematically perturbed counterparts—was embedded using the model’s internal representation space and projected into three principal components using PCA. The perturbations were designed to produce increasingly contradictory counter-knowledge (CK) statements, ranging from small to extreme deviations from the PM. Each sentence was then used in prompts to query the LLM, and the

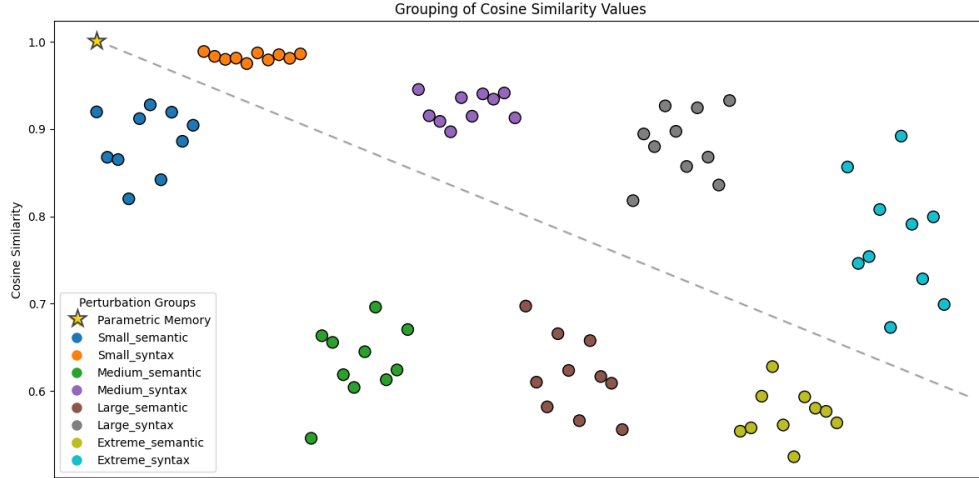


Figure 4.1: Cosine similarity values for counter knowledge generated off Llama-2-7B’s parametric memory. Above the dotted line are the cosine similarities of counter-knowledge instances that with altered syntax and consistent semantics. Below the dotted line are the inverse, where semantics stayed the same while words were changed. Neither alone led to natural-sounding language, prompting a unified approach.

resulting responses were labeled according to whether they aligned with the PM or with the counter-knowledge evidence.

The resulting figure reveals clear semantic clustering by perturbation strength: sentences within the same perturbation class occupy distinct and coherent regions in embedding space, indicating that the model’s representation captures the graded nature of the manipulations. Crucially, within each cluster, there is no clear separation between responses aligning with the PM and those aligning with the CK, suggesting that the model’s alignment behavior is not strongly reflected in embedding proximity. In other words, while the embeddings distinguish perturbation magnitude effectively, they do not encode whether a given prompt will elicit a memory-consistent or evidence-consistent answer.

We take this behavior to indicate that our natural-language perturbations, which vary both syntactically and semantically, are being interpreted by the model in structured and internally consistent ways. The resulting embedding clusters align with our expectations that as the degree of perturbation increases, the representations move progressively farther from the parametric knowledge. This pattern supports the idea that the model’s embedding space faithfully reflects the intended semantic continuum of our perturbation design. We hope these two findings help validate the utility of our methods over others for future research involving counter memory generation.

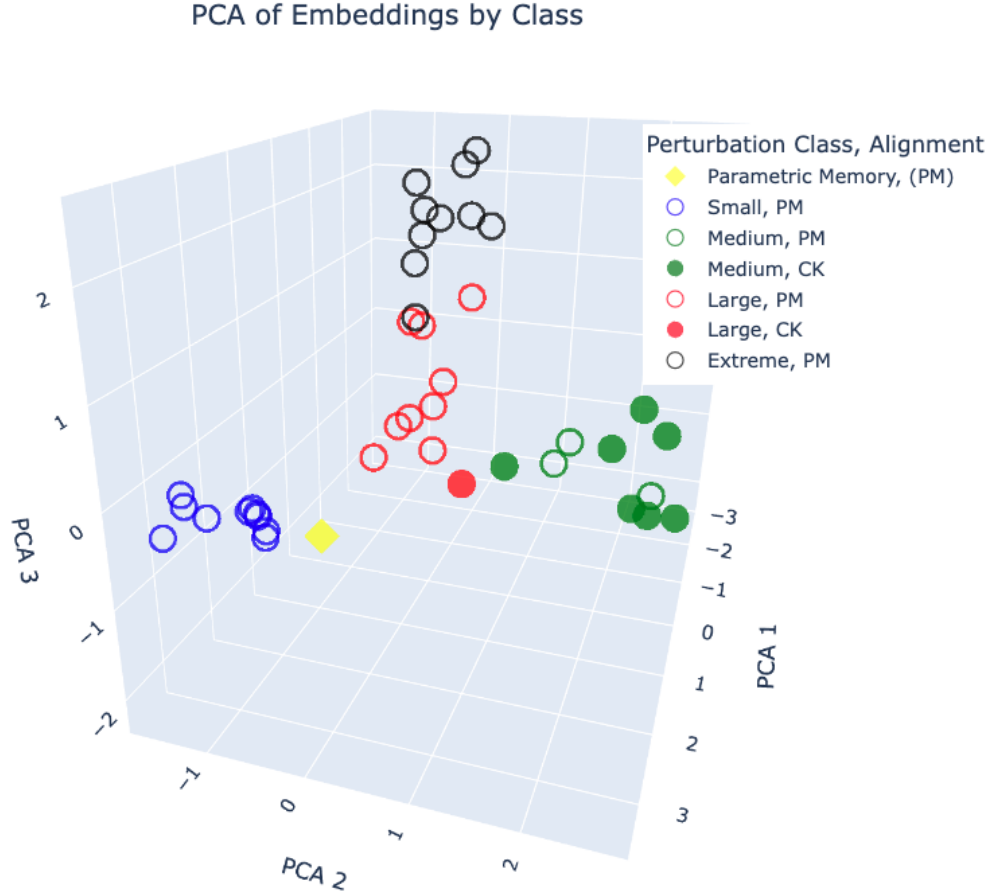


Figure 4.2: PCA visualization of sentence embeddings grouped by perturbation class and response alignment. Each point represents an embedded sentence projected into a 3D PCA space. The yellow diamond marks the Parametric Memory (PM) — the model’s original stored knowledge. Surrounding points are counter-knowledge (CK) sentences designed to increasingly contradict the PM across small, medium, large, and extreme perturbation levels. Hollow markers denote sentences for which the LLM’s answer aligned with the parametric memory, despite exposure to counter-knowledge. Filled markers indicate responses that aligned with the counter-knowledge evidence instead. The clustering shows how semantic distance from the PM relates to both perturbation strength and alignment behavior.

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